

Unit 10, Lesson 2: Solving One-Step Inequalities Using Addition and Subtraction

Benchmark

MA.7.AR.2.1 Write and solve one-step inequalities in one variable within a mathematical context and represent solutions algebraically or graphically.

Learning Targets

- ☐ I can solve one-step inequalities in one variable using addition or subtraction.
- ☐ I can relate properties of inequality to solving one-step inequalities algebraically.

10.2.1 Warm-Up: Cost Estimators

Cost estimators use data analytics to estimate the cost of manufacturing a product, constructing a building, or providing a service. They predict the amount of time, money, materials, and labor that will be needed to complete a project. This includes factoring in variables that may arise during the course of a project such as bad weather or shipping delays.

1. How do you think cost estimators use the minimum and maximum values of inequalities in their work?
2. Cost estimators usually specialize in a particular product or industry. Why do you think this is helpful?

10.2.2 Exploration: Properties of Inequality

Three inequalities are shown.

$$x > -2.3$$

$$-2.3 > x$$

$$-2.3 < x$$

1. Circle the inequalities that represent the same solution set.

2. Complete the statement.

For both inequalities, all the values of x that are _____ than _____ make both inequalities true.

These two inequalities are equivalent by the ***asymmetric property of inequality***.



The ***asymmetric property of inequality*** states that if $a > b$, then $b < a$.

3. Use the asymmetric property of inequality to write an inequality equivalent to $q + \frac{2}{5} \leq \frac{7}{10}$.
4. Four true inequalities are given in the table. Work with your partner to complete the table by following the steps indicated for each inequality and then answering the questions.

Given Inequality	Step	What is the new inequality?	Is the new inequality true?
$-6 < -2$	Add 8.1 to both sides		
$-5 < \frac{3}{4}$	Subtract 2 from both sides		
$7.45 > 7$	Add $3\frac{1}{5}$ to both sides		
$0 > -\frac{2}{3}$	Subtract $\frac{5}{6}$ from both sides		

5. Use your work on the previous problem to complete the description of each property.

Property of Inequality	Description
Addition property of inequality	When the same value is added to both sides of a true inequality, ...
Subtraction property of inequality	When the same value is subtracted from both sides of a true inequality, ...

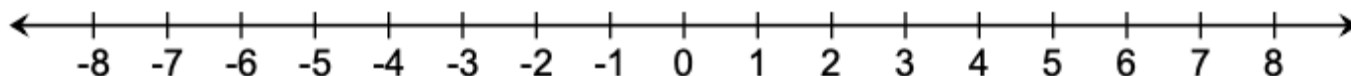
10.2.3 Guided Instruction: Solving One-Step Inequalities with Addition and Subtraction

1. Complete the table to determine the value of $x - 3$ for each value of x .

x	-4	-3.2	-2	$-\frac{1}{2}$	0	1	2	$3\frac{6}{7}$	4
$x - 3$	-7		-5				-1		1

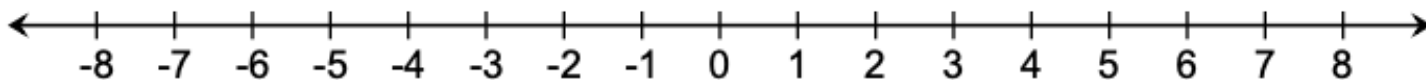
2. For which value of x is the equation $x - 3 = -2$ true?

3. Represent the solution to the equation $x - 3 = -2$ on the number line.



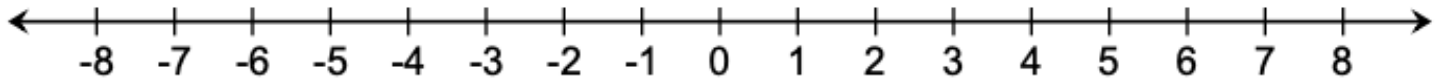
4. For which values of x in the table is the inequality $x - 3 > -2$ true?

5. Represent the values from the table that are part of the solution set of $x - 3 > -2$ on the number line.



The solution set of $x - 3 > -2$ includes more values than shown on the previous number line.

6. Graph the solution set for the inequality.



7. Write a value on the line to show an equivalent inequality represented by the graph above.

$$x > \underline{\hspace{1cm}}$$

The solution set of the inequality $x - 3 > -2$ is all the values of x that are greater than 1, or $x > 1$.

This solution set can be determined algebraically by adding the same value to both sides of the inequality.

8. What value can be added to both sides of the inequality $x - 3 > -2$ to isolate x ?

Adding the same value to both sides of an inequality can be used to solve an inequality by the ***addition property of inequality***.



Addition Property of Inequality

If $a > b$, then $a + c > b + c$.

The ***addition property of inequality*** states that if the same value is added to both sides of an inequality, the result will be an equivalent inequality that is also true.

The ***subtraction property of inequality*** can also be used to solve an inequality.



Subtraction Property of Inequality

If $a > b$, then $a - c > b - c$.

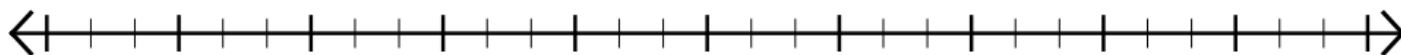
The ***subtraction property of inequality*** states that if the same value is subtracted from both sides of an inequality, the result will be an _____ inequality that is also _____.

These properties can be used to solve one-step inequalities algebraically.

9. Use the inequality $-12.25 > y - 2.4$ to complete the following.

a. Find the solution set by isolating the variable. Show your work.

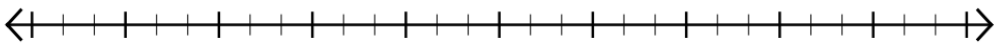
b. Graph the solution set on the number line.



c. Use substitution to test a value or values in order to verify that the solution and the graph are correct.

10.2.4 Guided Practice: Solving One-Step Inequalities with Addition and Subtraction

1. Use the inequality $-4\frac{3}{4} + c \leq -7.4$ to complete the following.
- a. Solve the inequality. Show your work.
- b. Graph the solution set on the number line.



- c. Use substitution to test a value or values to verify that the solution and the graph are correct.
2. Greg and Bianca both solved the inequality $-5.2 + x > -8.4$. Their solutions are shown.

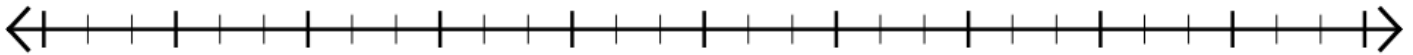
Greg’s Solution	Bianca’s Solution
$-3.2 < x$	$x < -3.2$

Explain who is correct. Show your work.

3. Michael got an estimate on the cost to repair his mountain bike. The parts will cost \$16.38 and he was told that the repair will be no more than \$40.50, including labor.

The inequality $16.38 + x \leq 40.50$ represents the situation where x is the cost of labor.

- a. Solve the inequality to determine the maximum amount the labor will cost.
- b. Graph the solution set on the number line below.



- c. Which solutions on the number line do not make sense for the context?

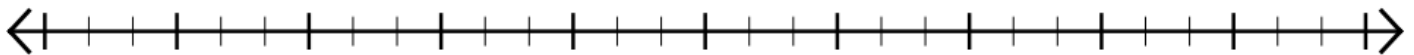
10.2.5 Your Turn!

Solve each inequality. Then, graph its solution set.

1. $y - 11 > -5$

a. Solve the inequality and show your work.

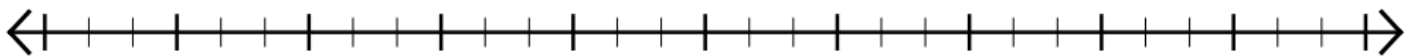
b. Graph the solution set.



2. $-8\frac{1}{2} \geq x + 7.25$

a. Solve the inequality and show your work.

b. Graph the solution set.



10.2.6 Wrap-Up: Cool Down

An equation and an inequality are shown.

$-2.25 = x + 2$

$-2.25 \leq x + 2$

1. Describe how the processes of solving the equation and the inequality are similar.
2. Complete the table to describe how the solutions to the equation and inequality are similar and different.

How are the solutions similar?	How are the solutions different?

